

Chapter 8

Systems of Linear 1st order DEs

8.1 Preliminary Theory

The following is an example of a system of linear 1st order DEs.

$$\begin{cases} \frac{dx}{dt} = 3x(t) + 2y(t) + \cos t \\ \frac{dy}{dt} = 5x(t) - 4y(t) + e^{at} \end{cases}$$

In matrix form, the system is

$$\frac{d}{dt} \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \underbrace{\begin{bmatrix} 3 & 2 \\ 5 & -4 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x(t) \\ y(t) \end{bmatrix}}_{\vec{X}(t)} + \underbrace{\begin{bmatrix} \cos t \\ e^{at} \end{bmatrix}}_{\vec{F}(t)}$$

In general, a system of n linear 1st order DEs has the form:

$$\frac{d}{dt} \vec{X}(t) = A \vec{X}(t) + \vec{F}(t)$$

Where A is an $n \times n$ matrix and $\vec{X}(t)$ and $\vec{F}(t)$ are column vectors in $\mathbb{R}^n$.

$\frac{d\vec{X}}{dt} = A\vec{X} + \vec{F}(t)$ is a nonhomogeneous [NH] system.

$\frac{d\vec{X}}{dt} = A\vec{X}$ is the corresponding homogeneous [H] system.

The general solution of $[NH]$ has the form :

$$\vec{X}(t) = \vec{X}_c(t) + \vec{X}_p(t)$$

where :

$\vec{X}_c(t)$ is the general solution of $[H]$

and

$\vec{X}_p(t)$ is a particular solution of $[NH]$.

If $\vec{X}_1(t), \vec{X}_2(t), \dots, \vec{X}_n(t)$ are linearly independent solutions of $[H]$, then:

$$\vec{X}_c(t) = c_1 \vec{X}_1(t) + c_2 \vec{X}_2(t) + \dots + c_n \vec{X}_n(t)$$

is the general solution of $[H]$ where

$c_1, c_2, \dots, c_n$ are arbitrary constants.

In section 8.2, we will learn how to solve

$$\frac{d\vec{X}}{dt} = A \vec{X}(t) \quad [H]$$

to find $\vec{X}_c(t)$. Eigenvalues and eigenvectors will be used.

In section 8.3, we will learn how to find a particular solution $\vec{X}_p(t)$ of

$$\frac{d\vec{X}}{dt} = A \vec{X}(t) + \vec{F}(t). \quad [NH]$$
